

Stereoselective synthesis of Xaa-Yaa type (Z)-chloroalkene dipeptide isosteres via efficient utilization of organocopper reagents mediated allylic alkylation



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ABSTRACT

An efficient method for an atom-economical synthesis of Xaa-Yaa type (Z)-chloroalkene dipeptide isosteres (CADIs) in high yield and diastereoselectivity has been developed. Reaction of an allylic *gem*-dichloride compound with organocopper reagents prepared from catalyst quantity of 30 mol% CuCl and organozinc reagents causes formation of (Z)-chloroalkenes corresponding to (L,D)-type Xaa-Yaa CADIs by diastereoselective allylic alkylation. Using (E)-enoates of substrates in place of (Z)-enoates in similar reactions with organocopper reagents, which prepared from 140 mol% CuCl and organozinc reagents, provides various (L,L)-type CADIs.

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1. Introduction

Peptides are important biomolecules with various bioactivities and a large number of peptides have been isolated and synthesized as lead compounds in drug discovery.¹ However, many bioactive peptides have low bioavailability because these peptides are easily decomposed by peptidases and the membrane permeability of hydrophilic peptides is limited. Backbone modification in a part or while of bioactive peptides as peptidomimetics is one of the most efficient methods with which to approach these problems.^{1c,2,3b}

Alkene dipeptide isosteres (ADIs), whose design is based on the partial double-bond character of native peptide bonds in the ground state conformation, have been thought to be ideal amide bond mimetics in dipeptides. Many groups have tried to convert amide bonds in peptides to produce normal (E)-ADIs (EADIs),^{3,4b,4d} fluoroalkene dipeptide isosteres (FADIs)⁴ or other type of dipeptide isosteres.⁵ However, bioactive peptides containing ADIs have not always been effective as dipeptide mimetics because, as a result of electronegativity, conventional ADIs could possess a smaller dipole moment. Consequently, these ADIs lack the steric restriction between the carbonyl oxygen and the side chain on the β -carbon in a native peptide bond due to the Van der Waals radius (VDR) which is

smaller than that of the original amide bond. In addition, various ADIs are not always accessible.

Recently, we have focused on chloroalkene structures,⁶ which are available versatile synthetic intermediates⁷ found in several marine natural products,⁸ and applied this to the synthesis of chloroalkene dipeptide isosteres (CADIs), which are based on the replacement of an amide bond in peptides with a chloroalkene (Fig. 1). CADIs can be expected to be more ideal peptide bond mimetics because the carbonyl oxygen atom of the amide bond is replaced by a chlorine atom, with similar electronegativity. We conceived that a chloro-olefin conformation might mimic the steric restriction experienced by the amide bond due to a chlorine atom which is larger than a carbonyl oxygen atom (Fig. 2).

In this way, we expected to overcome the drawbacks of ADIs. Few examples however have been reported of applications of chloroalkene moieties as peptidomimetics.⁹ Therefore, in this paper we report an expansion of our earlier work on the development of a stereoselective synthetic method for various Xaa-Yaa type CADIs. Utilization of catalyst or equivalent quantity of copper salt (I) is investigated to proceed the reaction efficiently. In addition, the conversion of CADIs to Boc- or Fmoc-protected carboxylic acids is performed for use in liquid- or solid-phase peptide synthesis.

2. Results and discussion

Stereoselective formation of functionalized alkenes is important

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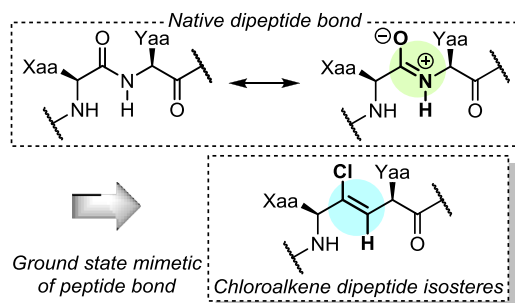


Fig. 1. A native dipeptide bond and designed chloroalkene dipeptide isosteres.

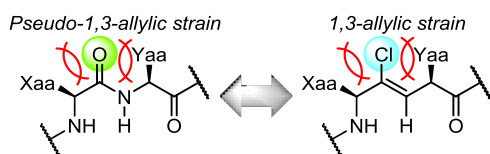
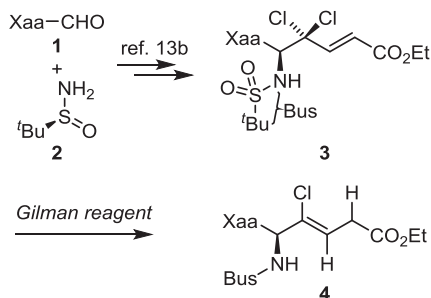


Fig. 2. Mimetic of pseudo-1,3-allylic strain.

because it enables control of functional groups or chiral positions for the synthesis of intermediate compounds, natural products or functional molecules.¹⁰ Stereoselective allylic substitution of organometallic reagents is a powerful tool for carbon–carbon bond formation in organic synthesis,¹¹ and allylic alkylation utilizing organocopper reagents is an efficient method for introduction of various alkyl groups.¹²

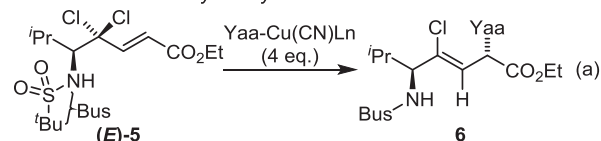
We previously reported an efficient synthetic method utilizing organocopper reagents. Treatment of the chloroalkene precursor (2*E*)- γ,γ -dichloro- α,β -enoate **3**¹³ which was prepared from the aldehyde **1** of the corresponding alkyl side chain of an amino acid and (S)-*tert*-butylsulfonamide **2**,¹⁴ with Gilman's reagent provided Bus-L-Xaa- $\psi[(Z)\text{-CCl=CH}]\text{-Gly-OEt} **4** by carbon–chlorine bond cleavage via a single electron transfer (SET) mechanism (Scheme 1).^{13a,b} The α -alkylated product (Bus-L-Val- $\psi[(Z)\text{-CCl=CH}]\text{-D-Yaa-OEt} **6** was obtained by remote stereoinduction in the organocuprate-mediated diastereoselective allylic alkylation of the substrate (*E*)-**5** (Scheme 2a).^{13a}$$

In addition, treatment of the substrate (*Z*)-**5**, which has the opposite olefin geometry, with cyanocuprates obtained (L,L)-type CADIs (Bus-L-Val- $\psi[(Z)\text{-CCl=CH}]\text{-L-Yaa-OEt}) in high yield and excellent diastereoselectivity (Scheme 2b).^{13a} These synthetic strategies can be applied to diastereoselective syntheses of (L,L), (L,D), (D,L), and (D,D)-type dipeptide isosteres by using an appropriate aldehyde and chiral sulfonamide as starting compounds, and$

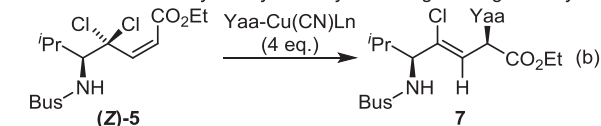


Scheme 1. Reductive dechlorination of γ,γ -dichloro- α,β -enoate **3** with Gilman's reagent.

Diastereoselective allylic alkylation via remote stereoinduction



Diastereoselective allylic alkylation by switching olefin geometry



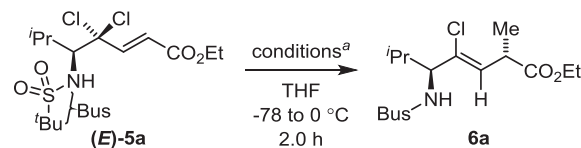
Scheme 2. Diastereoselective allylic alkylation utilizing organocopper reagents.

an intermediate of an (*E*)- or (*Z*)-enoate. By introduction at a subsequent stage of various alkyl groups at the α -position, these can contribute to a diversity-oriented synthesis of (*Z*)-chloroalkene peptide isosteres.

Seeking an efficient synthesis of CADIs, we investigated decreasing the amount of copper salt required for the preparation of organocopper reagents (Table 1). Treatment of (*E*)-**5** with 50 mol % of CuCN or CuI and 4 equivalents of the organozinc reagent afforded the diastereoselective allylic alkylated product **6a** in sufficient yield and diastereoselectivity (entries 1–2). Even decreasing to the amount of copper salts by up to 20 mol% provided similar results (Table 1, entries 3–5). The alkylated compound **6a** was also obtained by using different copper salts such as CuBr, CuCl, Cu(OAc), and Cu(TC) (entries 6–9). CuCl was the best among the copper salts used (entry 7). Evaluation of the quantity of CuCl used showed that 30 mol% CuCl was provided excellent yield and diastereoselectivity (entry 10). Decreasing the amount of the organozinc reagent by up to 2.4 equivalents gave the same result (entry 11). Based on these results, the condition of 30 mol% CuCl (entry 11) was adopted as the best condition of this reaction. Furthermore, these results showed that copper salts functioned as catalysts in the organozinc-copper-mediated allylic alkylation.

Table 1

Reactivity of (*E*)-**5** with organocopper reagents prepared by different amount of copper salts.



entry	copper salt	6a , Z/E ^c	6a , dr ^c	6a , yield ^d (%)
1	50 mol% CuCN	>20:1	>20:1	83
2	50 mol% CuI	>20:1	>20:1	94
3	30 mol% CuI	>20:1	>20:1	93
4	20 mol% CuI	>20:1	>20:1	80
5	10 mol% CuI	>20:1	>20:1	63
6	20 mol% CuBr	>20:1	>20:1	73
7	20 mol% CuCl	>20:1	>20:1	88
8	20 mol% Cu(OAc)	>20:1	>20:1	78
9	20 mol% Cu(TC)	>20:1	>20:1	78
10	30 mol% CuCl	>20:1	>20:1	99
11 ^b	30 mol% CuCl	>20:1	>20:1	99

^aAll reactions were carried out at -78°C for 1.0 h on a 0.3 mmol scale with 4.0 equiv of MeZnCl and 8.0 equiv of LiCl.

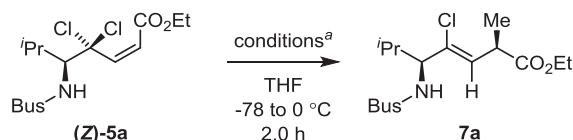
^b With 2.4 equiv of MeZnCl and 6.0 equiv of LiCl.

^c Determined by ^1H NMR of unpurified reaction mixture.

^d Yields were determined by ^1H NMR of unpurified reaction mixture using 1,4-dinitrobenzene as an internal standard.

Table 2

Reactivity of (**Z**)-**5** with organocopper reagents prepared by different amount of copper salts.



entry	copper salt	7a , Z/E ^b	7a , dr ^b	7a , yield ^c (%)
1	50 mol% CuCN	>20:1	>20:1	5.7
2	30 mol% CuCN	>20:1	>20:1	11
3	50 mol% CuCl	>20:1	>20:1	48
4	100 mol% CuCl	>20:1	>20:1	92
5	140 mol% CuCl	>20:1	>20:1	99

^a All reactions were carried out at -78°C for 1.0 h on a 0.3 mmol scale with 2.4 equiv of MeZnCl and 6.0 equiv of LiCl.

^b Determined by ^1H NMR of unpurified reaction mixture.

^c Yields were determined by ^1H NMR of unpurified reaction mixture using 1,4-dinitrobenzene as an internal standard.

Since diastereoselective allylic alkylation using copper salts and organozinc reagents was successfully achieved, we have considered preparing (L,L)-type CADIs (Bus-L-Val-ψ[(Z)-CCl=CH]-L-Yaa-OEt) by switching the olefin geometry of the substrates from an (*E*)-enoate to a (*Z*)-enoate with a small amount of copper salt. Thus, (*Z*)-enoate (**Z**)-**5** was prepared and used in the allylic alkylation (Table 2). Although the reactions with catalytic amounts of copper salts gave low yields of the α-alkylated product (entries 1–3), the use of an equivalent of both copper salts and organozinc reagents provided compound **7a** in high yield (entry 4). In case of the substrate (**Z**)-**5**, since a copper salt was trapped by the alkylated intermediate, more amount of the copper salt was required and it was difficult to work as a catalyst. Therefore, treatment of the substrate (**Z**)-**5** with 140 mol% CuCl and MeZnCl led to excellent conversion and

diastereoselectivity (entry 5) and we selected the 140 mol% CuCl as the optimal condition.

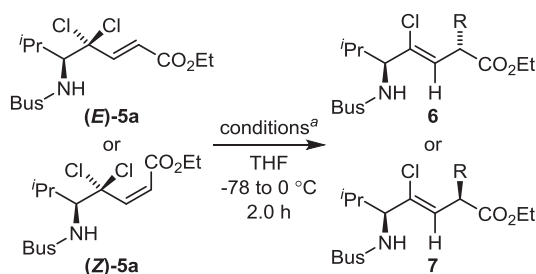
With the optimal conditions for providing (L,D)- or (L,L)-type CADIs identified, the scope of this diastereoselective allylic alkylation was explored (Table 3). Treatment of the substrate with the organocopper reagents, which were prepared from the 30 mol% or 140 mol% of the copper catalyst and organozinc reagents respectively, provided the corresponding α-alkylated products (**6a–e**, **7a–e**) in high yield and diastereoselectivity (entries 1–10). In particular, the lower levels of copper reagent afforded α-allylation products in higher yield than the cyanocuprate (entries 5, 10).^{13a,15}

To investigate whether various α-alkylated products could be produced with decreased levels of copper salts, we explored the scope of substrates in this diastereoselective allylic alkylation. Treatment of various (*E*)- or (*Z*)-enoates under optimized conditions led to the corresponding alkylated compounds (Table 4). In all cases, the diastereoselective allylic alkylation products (**9a–c**, **10a–c**) were produced with excellent Z-selectivity (>20:1). The substrates with sterically bulky alkyl groups or an aromatic group also produced alkylated products in excellent yields and diastereoselectivity (>20:1).

The synthetic methods for various (L,D)- or (L,L)-type CADIs having been established, we attempted to convert a CADI **7c** into a Boc-protected carboxylic acid (Boc-L-Val-ψ[(Z)-CCl=CH]-L-Phe-OH) **12** or an Fmoc-protected compound (Fmoc-L-Val-ψ[(Z)-CCl=CH]-L-Phe-OH) **14** (Scheme 3). Deprotection of the Bus group of **7c** with AlCl_3 and anisole¹⁶ followed by Boc- or Fmoc- protection, produced the Boc-protected ester **11** or the Fmoc-protected ester **13**. Finally, the ester group of **11** or **13** was reduced to the corresponding aldehyde at -78°C and this was followed by Pinnick oxidation to provide the desired Boc-protected carboxylic acid (Boc-L-Val-ψ[(Z)-CCl=CH]-L-Phe-OH) **12** or Fmoc-protected compound (Fmoc-L-Val-ψ[(Z)-CCl=CH]-L-Phe-OH) **14** in high overall yield from **7c** without epimerization or olefin isomerization.

Table 3

Scope of diastereoselective allylic alkylation of (*E*)- and (*Z*)-**5a**.

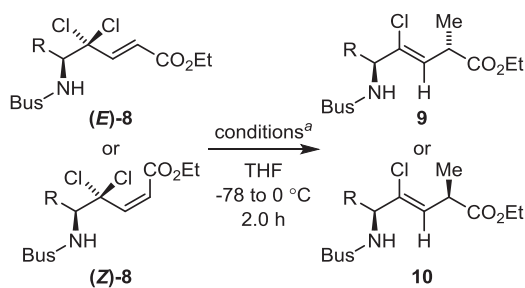


entry	substrate	copper salt	RZnX	Z/E ^b	dr ^b	6 or 7 , yield ^c (%)
1	(<i>E</i>)- 5	30 mol% CuCl	MeZnCl	>20:1	>20:1	6a , 99
2	(<i>E</i>)- 5	30 mol% CuCl	ⁱ BuZnBr	>20:1	>20:1	6b , 98
3	(<i>E</i>)- 5	30 mol% CuCl	BnZnBr	>20:1	>20:1	6c , 97
4	(<i>E</i>)- 5	30 mol% CuCl	$\text{EtO}_2\text{C}(\text{CH}_2)_2\text{ZnBr}$	>20:1	>20:1	6d , 98
5	(<i>E</i>)- 5	30 mol% CuCl	AllylZnCl	>20:1	>20:1	6e , 70
6	(<i>Z</i>)- 5	140 mol% CuCl	MeZnCl	>20:1	>20:1	7a , 97
7	(<i>Z</i>)- 5	140 mol% CuCl	ⁱ BuZnBr	>20:1	>20:1	7b , 95
8	(<i>Z</i>)- 5	140 mol% CuCl	BnZnBr	>20:1	>20:1	7c , 94
9	(<i>Z</i>)- 5	140 mol% CuCl	$\text{EtO}_2\text{C}(\text{CH}_2)_2\text{ZnBr}$	>20:1	>20:1	7d , 96
10	(<i>Z</i>)- 5	140 mol% CuCl	AllylZnCl	>20:1	>20:1	7e , 66

^a All reactions were carried out at -78°C for 30 min on a 0.3 mmol scale with 30 or 140 mol% of CuCl and 2.4 equiv of organozinc reagent in the presence of 6.0 equivalent of LiCl.

^b Determined by ^1H NMR of an unpurified reaction mixture.

^c Yields of isolated products.

Table 4Scope of substrates (**E**)-**8** and (**Z**)-**8** in diastereoselective allylic alkylation.

entry	substrate	R	copper salt	Z/E ^b	dr ^b	9 or 10 , yield ^c (%)
1	(E)- 8a	ⁱ Bu	30 mol% CuCl	>20:1	>20:1	9a , 93
2	(E)- 8b	ⁿ Bu	30 mol% CuCl	>20:1	>20:1	9b , 98
3	(E)- 8c	Bn	30 mol% CuCl	>20:1	>20:1	9c , 95
4	(Z)- 8a	ⁱ Bu	140 mol% CuCl	>20:1	>20:1	10a , 96
5	(Z)- 8b	ⁿ Bu	140 mol% CuCl	>20:1	>20:1	10b , 97
6	(Z)- 8c	Bn	140 mol% CuCl	>20:1	>20:1	10c , 96

^aAll reactions were carried out at -78 °C for 30 min on a 0.3 mmol scale with 30 or 140 mol% of CuCl and 2.4 equiv of MeZnCl in the presence of LiCl.

^bDetermined by ¹H NMR of an unpurified reaction mixture.

^cYields of isolated products.

3. Conclusion

In conclusion, we have developed an atom-economical synthesis of (L,D)- or (L,L)-type CADIs utilizing diastereoselective allylic alkylation. Treatment of the substrates in accompany with switching the olefin geometry of allylic *gem*-dichlorides using 30 mol% or 140 mol% CuCl and organozinc reagents has provided

various (L,D)- or (L,L)-type CADIs. In this conditions, allylic alkylated products were obtained by small amount of copper salt (**1**) in high yield and excellent diastereoselectivity. Since reaction of γ,γ -dichloro- α,β -enoates was performed by 30 mol% CuCl and organozinc reagents, it was proven that copper salts functioned as catalysts in the organozinc-copper-mediated allylic alkylation. In addition, a Boc- or Fmoc-protected carboxylic acid can be easily prepared for liquid- or solid-phase peptide synthesis from a common intermediate in a few steps and with high total yield. These synthetic methodologies are a model for production of various Xaa-Yaa type dipeptide isosteres. Currently, the synthesis and evaluation of bioactive peptides containing CADIs is in progress.

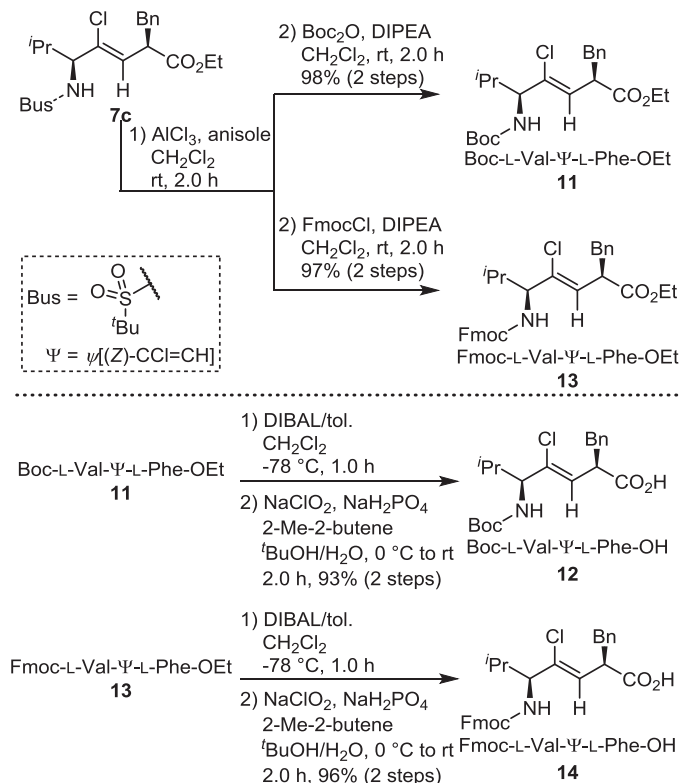
4. Experimental section

4.1. General methods

All reactions using air- or moisture-sensitive reagents were performed in dried glassware under an atmosphere of nitrogen, with commercially supplied solvents and reagents unless otherwise noted. CH₂Cl₂ was distilled from CaH₂ and stored over molecular sieves 4A. Thin-layer chromatography (TLC) was performed on Merck 60F₂₅₄ precoated silica gel plates and was visualized by fluorescence quenching under UV light and by staining with phosphomolybdic acid, *p*-anisaldehyde, or ninhydrin. Flash column chromatography was carried out silica gel 60 N (Kanto Chemical Co., Inc.).

4.2. Characterization

¹H NMR (400 or 500 MHz) and ¹³C NMR (100 or 125 MHz) spectra were recorded using a Bruker Avance II spectrometer. Chemical shifts are reported in δ (ppm) relative to Me₄Si (in CDCl₃)



Scheme 3. Conversion into Boc- or Fmoc-L-Val-Ψ[(Z)-CCl=CH]-L-Phe-OH (**12**, **14**).

as internal standard. Infrared (IR) spectra were recorded on a JASCO FT/IR 4100, and are reported in wavenumbers (cm^{-1}). Low- and high-resolution mass spectra were recorded on a Bruker Daltonics micrOTOF focus (ESI-MS) spectrometers in positive and negative detection modes.

Optical rotations were measured on a JASCO DIP-370 polarimeter operating at the sodium D line with a 100 mm path length cell, and were reported as follows: $[\alpha]_D$ (concentration (g/100 mL), solvent).

4.3. Experimental procedures for synthesis of compounds

4.3.1. Ethyl (*S,Z*)-4,4-dichloro-5-((1,1-dimethylethyl)sulfonamido)-7-methyloct-2-enoate (**(Z)-8a**)

A solution of DIBAL-H in toluene (1.00 M, 1.50 mL, 1.50 mmol) was added dropwise to a solution of the previously reported methyl ester^{13b} (348 mg, 4.10 mmol) in CH_2Cl_2 (10.0 mL) at -78°C under N_2 , and the mixture was stirred at -78°C for 2.0 h. The reaction was quenched with saturated aqueous Rochelle salt solution and extracted with Et_2O . The extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure gave an oily aldehyde, which was used immediately in the next step without purification. NaI (172 mg, 1.15 mmol) and DBU (157 μL , 1.05 mmol) were added to a stirred solution of ethyl di-*o*-tolylphosphonoacetate¹⁷ (307 μL , 1.05 mmol) in THF (9.00 mL) at 0°C for 10 min under N_2 . After the mixture was cooled up to -78°C , a solution of the above aldehyde in THF (1.00 mL) was added. After the mixture was stirred at -78°C for 10 min, the resulting mixture was slowly warmed up to 0°C . The reaction was quenched with saturated aqueous NH_4Cl solution and extracted with EtOAc . The extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane- EtOAc (20:1) gave the title compound (**(Z)-8a**) as a colorless oil (286 mg, 74%): $[\alpha]_D = -24.0$ (c 1.04, CHCl_3); IR (ATR) ν 1727 (CO), 1318 (NHSO₂), 1126 (NHSO₂) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 1.00–1.04 (m, 6H), 1.29–1.32 (m, 3H), 1.50 (s, 9H), 1.52–1.60 (m, 1H), 1.89–2.00 (m, 2H), 3.16–3.19 (m, 1H), 4.17–4.24 (m, 2H), 5.95 (d, $J = 11.5$ Hz, 1H), 6.99 (d, $J = 11.5$ Hz, 1H); ^{13}C NMR (125 MHz, CDCl_3) δ 1.78, 14.3, 22.4, 23.2, 24.0 (2C), 26.0, 38.3, 49.4, 60.5, 60.9, 66.2, 122.0, 138.0, 164.0; HRMS (ESI), m/z calcd for $\text{C}_{15}\text{H}_{28}\text{Cl}_2\text{NO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 388.1111, found 388.1113.

4.3.2. Ethyl (*S,Z*)-4,4-dichloro-5-((1,1-dimethylethyl)sulfonamido)non-2-enoate (**(Z)-8b**)

A solution of DIBAL-H in toluene (1.00 M, 1.50 mL, 1.50 mmol) was added dropwise to a solution of the reported methyl ester^{13b} (348 mg, 1.00 mmol) in CH_2Cl_2 (10.0 mL) at -78°C under N_2 , and the mixture was stirred at -78°C for 2.0 h. The reaction was quenched with saturated aqueous Rochelle salt solution and extracted with Et_2O . The extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure gave an oily aldehyde, which was used immediately in the next step without purification. NaI (172 mg, 1.15 mmol) and DBU (157 μL , 1.15 mmol) were added to a stirred solution of ethyl di-*o*-tolylphosphonoacetate¹⁷ (307 μL , 1.05 mmol) in THF (9.00 mL) at 0°C for 10 min under N_2 . After the mixture was cooled to -78°C , a solution of the above aldehyde in THF (1.00 mL) was added. The mixture was stirred at -78°C for 10 min, then was slowly warmed up to 0°C . The reaction was quenched with saturated aqueous NH_4Cl solution and extracted with EtOAc . The extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane- EtOAc (30:1) gave the title compound (**(Z)-8b**) as a colorless oil (321 mg, 83%): $[\alpha]_D = -35.0$ (c 1.05, CHCl_3); IR (ATR) ν 1726 (CO), 1318 (NHSO₂), 1125 (NHSO₂) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3)

δ 0.929–0.959 (m, 3H), 1.30 (t, $J = 7.15$ Hz, 3H), 1.38–1.72 (m, 15H), 2.07–2.13 (m, 1H), 4.21 (q, $J = 7.15$ Hz, 2H), 5.95 (d, $J = 11.5$ Hz, 1H), 6.99 (d, $J = 11.5$ Hz, 1H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.0, 14.3, 22.7 (3C), 24.0, 28.6, 29.0, 51.0, 60.5, 60.9, 66.1, 122.0, 138.0, 164.0; HRMS (ESI), m/z calcd for $\text{C}_{15}\text{H}_{28}\text{Cl}_2\text{NO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 388.1111, found 388.1115.

4.3.3. Ethyl (*S,Z*)-4,4-dichloro-5-((1,1-dimethylethyl)sulfonamido)-6-phenylhex-2-enoate (**(Z)-8c**)

A solution of DIBAL-H in toluene (1.00 M, 1.50 mL, 1.50 mmol) was added dropwise to a solution of the reported methyl ester^{13b} (382 mg, 1.00 mmol) in CH_2Cl_2 (41.0 mL) at -78°C under N_2 , and the mixture was stirred at -78°C for 2.0 h. The reaction was quenched with saturated aqueous Rochelle salt solution and extracted with Et_2O . The extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure gave an oily aldehyde, which was used immediately in the next step without purification. NaI (172 mg, 1.15 mmol) and DBU (157 μL , 1.05 mmol) were added to a stirred solution of ethyl di-*o*-tolylphosphonoacetate¹⁷ (307 μL , 1.05 mmol) in THF (9.00 mL) at 0°C for 10 min under N_2 . After the mixture was cooled up to -78°C , a solution of the above aldehyde in THF (1.00 mL) was added. After the mixture was stirred at -78°C for 10 min, the resulting mixture was slowly warmed up to 0°C . The reaction was quenched by saturated aqueous NH_4Cl solution and extracted with EtOAc . The extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane- EtOAc (20:1) gave the title compound (**(Z)-8c**) as a colorless oil (384 mg, 91%): $[\alpha]_D = -14.0$ (c 1.00, CHCl_3); IR (ATR) ν 1722 (CO), 1318 (NHSO₂), 1127 (NHSO₂) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 1.25 (s, 9H), 1.30 (t, $J = 7.15$ Hz, 3H), 2.88 (dd, $J = 15.2, 8.6$ Hz, 1H), 3.46 (dd, $J = 8.6, 3.00$ Hz, 1H), 3.67 (dd, $J = 15.2, 3.00$ Hz, 1H), 4.20 (q, $J = 7.15$ Hz, 2H), 5.99 (d, $J = 11.4$ Hz, 1H), 7.11 (d, $J = 11.4$ Hz, 1H), 7.12–7.33 (m, 5H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.3, 23.8 (3C), 35.7, 52.5, 60.9, 61.0, 64.9, 123.0, 127.0, 129.0 (2C), 130.0 (2C), 137.0, 138.0164.0; HRMS (ESI), m/z calcd for $\text{C}_{18}\text{H}_{26}\text{Cl}_2\text{NO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 422.0954, found 422.0956.

4.3.4. Ethyl (2*S*,5*S*,*Z*)-4-chloro-5-((1,1-dimethylethyl)sulfonamido)-2,7-dimethyloct-3-enoate (**9a**)

A solution of methylzinc chloride in THF (2.00 M, 360 μL , 0.720 mmol) was added to a suspension of CuCl (8.91 mg, 90.0 μmol) and LiCl (76.3 mg, 1.80 mmol) in THF (2.0 mL) at -78°C under N_2 , and the mixture was stirred at 0°C for 10 min. A solution of (**E**)-**8a**^{13b} (117 mg, 0.300 mmol) in THF (1.00 mL) was added dropwise to the solution of the organocopper reagent at -78°C . After being stirred at 0°C for 2.0 h, the reaction was quenched by addition of NH_4Cl -28% NH_3 aqueous solution with additional stirring at room temperature for 30 min. The mixture was extracted with Et_2O , and the extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane- EtOAc (7:1) gave the title compound **9a** as a colorless oil (102 mg, 93%, >20:1 dr): $[\alpha]_D = +38.6$ (c 0.990, CHCl_3); IR (ATR) ν 1735 (CO), 1367 (NHSO₂), 1127 (NHSO₂) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 0.924–0.965 (m, 6H), 1.25 (d, $J = 7.15$ Hz, 3H), 1.27 (t, $J = 7.10$ Hz, 3H), 1.36 (s, 9H), 1.54–1.62 (m, 4H), 3.57 (dq, $J = 9.05, 7.15$ Hz, 1H), 4.14 (q, $J = 7.10$ Hz, 2H), 4.18–4.23 (m, 1H), 5.86 (d, $J = 9.05$ Hz, 1H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.3, 17.2, 22.1, 22.7, 24.3 (3C), 24.8, 39.6, 44.6, 58.6, 59.8, 61.1, 127.0, 136.0, 173.0; HRMS (ESI), m/z calcd for $\text{C}_{16}\text{H}_{31}\text{ClNO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 368.1657, found 368.1660.

4.3.5. Ethyl (2*S*,5*S*,*Z*)-4-chloro-5-((1,1-dimethylethyl)sulfonamido)-2-methylnon-3-enoate (**9b**)

A solution of methylzinc chloride in THF (2.00 M, 360 μL ,

0.720 mmol) was added to a suspension of CuCl (8.91 mg, 90.0 μ mol) and LiCl (76.3 mg, 1.80 mmol) in THF (2.0 mL) at -78°C under N_2 , and the mixture was stirred at 0°C for 10 min. A solution of (**E**)-**8b**^{13b} (117 mg, 0.300 mmol) in THF (1.00 mL) was added dropwise to the solution of the organocopper reagent at -78°C . After stirring at 0°C for 2.0 h, the reaction was quenched by addition of NH_4Cl -28% NH_3 aqueous solution with additional stirring at room temperature for 30 min. The mixture was extracted with Et_2O , and the extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane-EtOAc (7:1) gave the title compound **9b** as a colorless oil (108 mg, 98%, >20:1 dr): $[\alpha]_{\text{D}} = +41.6$ (c 1.02, CHCl_3); IR (ATR) ν 1738 (CO), 1367 (NHSO_2), 1130 (NHSO_2) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 0.887–0.915 (m, 3H), 1.24–1.36 (m, 19H), 1.68–1.73 (m, 2H), 3.55–3.61 (m, 1H), 4.06–4.11 (m, 2H), 4.11–4.16 (m, 2H), 5.83–5.85 (m, 1H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.0, 14.3, 17.2, 22.2, 24.3 (3C), 28.1, 35.2, 39.6, 59.8, 60.3, 61.1, 127.0, 136.0, 174.0; HRMS (ESI), m/z calcd for $\text{C}_{16}\text{H}_{31}\text{ClNO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 368.1657, found 368.1659.

4.3.6. Ethyl 2*S*,5*S*,*Z*)-4-chloro-5-((1,1-dimethylethyl)sulfonamido)-2-methyl-6-phenylhex-3-enoate (**9c**)

A solution of methylzinc chloride in THF (2.00 M, 360 μL , 0.720 mmol) was added to a suspension of CuCl (8.91 mg, 90.0 μ mol) and LiCl (76.3 mg, 1.80 mmol) in THF (2.0 mL) at -78°C under N_2 , and the mixture was stirred at 0°C for 10 min. A solution of (**E**)-**8c** (127 mg, 0.300 mmol) in THF (1.00 mL) was added dropwise to the solution of the organocopper reagent at -78°C . After stirring at 0°C for 2.0 h, the reaction was quenched by addition of NH_4Cl -28% NH_3 aqueous solution with additional stirring at room temperature for 30 min. The mixture was extracted with Et_2O , and the extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane-EtOAc (7:1) gave the title compound **9c** as a light yellow oil (114 mg, 95%, >20:1 dr): $[\alpha]_{\text{D}} = +36.1$ (c 1.03, CHCl_3); IR (ATR) ν 1736 (CO), 1367 (NHSO_2), 1130 (NHSO_2) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 0.961 (d, $J = 7.15$ Hz, 3H), 1.21 (t, $J = 7.10$ Hz, 3H), 1.32 (s, 9H), 2.97 (dd, $J = 13.2$, 8.90 Hz, 1H), 3.12 (dd, $J = 13.2$, 5.10 Hz, 1H), 3.45 (dq, $J = 9.20$, 7.15 Hz, 1H), 4.08 (q, $J = 7.10$ Hz, 2H), 4.26 (d, $J = 9.50$ Hz, 1H), 4.29 (ddd, $J = 9.50$, 8.90, 5.10 Hz, 1H), 5.47 (d, $J = 9.20$ Hz, 1H), 7.16–7.28 (m, 5H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.3, 16.7, 24.2 (3C), 39.4, 41.8, 60.1, 61.0, 62.1, 127.0, 128.0, 129.0 (2C), 130.0 (2C), 134.0, 136.0, 173.0; HRMS (ESI), m/z calcd for $\text{C}_{19}\text{H}_{29}\text{ClNO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 402.1500, found 402.1502.

4.3.7. Ethyl (2*R*,5*S*,*Z*)-4-chloro-5-((1,1-dimethylethyl)sulfonamido)-2,7-dimethyloct-3-enoate (**10a**)

A solution of methylzinc chloride in THF (2.00 M, 360 μL , 0.720 mmol) was added to a suspension of CuCl (41.6 mg, 0.420 mmol) and LiCl (76.3 mg, 1.80 mmol) in THF (2.00 mL) at -78°C under N_2 , and the mixture was stirred at 0°C for 10 min. A solution of (**Z**)-**8a** (117 mg, 0.300 mmol) in THF (1.00 mL) was added dropwise to the solution of this organocopper reagent at -78°C . After being stirred at 0°C for 2.0 h, the reaction was quenched by addition of NH_4Cl -28% NH_3 aqueous solution with additional stirring at room temperature for 30 min. The mixture was extracted with Et_2O , and the extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane-EtOAc (8:1) gave the title compound **10a** as a colorless oil (106 mg, 96%, >20:1 dr): $[\alpha]_{\text{D}} = -23.6$ (c 0.990, CHCl_3); IR (ATR) ν 1734 (CO), 1367 (NHSO_2), 1127 (NHSO_2) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 0.915–0.956 (m, 6H), 1.25 (t, $J = 7.10$ Hz, 3H), 1.28 (d, $J = 7.20$ Hz, 3H), 1.37 (s, 9H), 1.50–1.65 (m, 3H), 3.58 (dq, $J = 9.00$, 7.20 Hz, 1H),

4.00–4.02 (m, 1H), 4.14 (q, $J = 7.10$ Hz, 2H), 4.19–4.24 (m, 1H), 5.86 (d, $J = 9.00$ Hz, 1H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.3, 16.9, 22.0, 22.8, 24.3 (3C), 24.6, 39.6, 44.2, 58.7, 59.8, 61.1, 127.0, 136.0, 174.0; HRMS (ESI), m/z calcd for $\text{C}_{16}\text{H}_{31}\text{ClNO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 368.1657, found 368.1661.

4.3.8. Ethyl (2*R*,5*S*,*Z*)-4-chloro-5-((1,1-dimethylethyl)sulfonamido)-2-methylnon-3-enoate (**10b**)

A solution of methylzinc chloride in THF (2.00 M, 360 μL , 0.720 mmol) was added to a suspension of CuCl (41.6 mg, 0.420 mmol) and LiCl (76.3 mg, 1.80 mmol) in THF (2.00 mL) at -78°C under N_2 , and the mixture was stirred at 0°C for 10 min. A solution of (**Z**)-**8b** (117 mg, 0.300 mmol) in THF (1.00 mL) was added dropwise to the solution of the organocopper reagent at -78°C . After stirring at 0°C for 2.0 h, the reaction was quenched by addition of NH_4Cl -28% NH_3 aqueous solution with additional stirring at room temperature for 30 min. The mixture was extracted with Et_2O , and the extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane-EtOAc (7:1) gave the title compound **10b** as a colorless oil (108 mg, 97%, >20:1 dr): $[\alpha]_{\text{D}} = -20.4$ (c 1.00, CHCl_3); IR (ATR) ν 1734 (CO), 1366 (NHSO_2), 1127 (NHSO_2) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 0.883–0.912 (m, 3H), 1.24–1.39 (m, 19H), 1.67–1.72 (m, 2H), 3.55–3.61 (m, 1H), 4.04–4.05 (m, 1H), 4.09–4.17 (m, 3H), 5.83–5.85 (m, 1H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.0, 14.3, 17.0, 22.3, 24.3 (3C), 28.0, 34.9, 39.6, 59.8, 60.4, 61.1, 127.0, 136.0, 174.0; HRMS (ESI), m/z calcd for $\text{C}_{16}\text{H}_{31}\text{ClNO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 368.1657, found 368.1659.

4.3.9. Ethyl 2*R*,5*S*,*Z*)-4-chloro-5-((1,1-dimethylethyl)sulfonamido)-2-methyl-6-phenylhex-3-enoate (**10c**)

A solution of methylzinc chloride in THF (2.00 M, 360 μL , 0.720 mmol) was added to a suspension of CuCl (41.6 mg, 0.420 mmol) and LiCl (76.3 mg, 1.80 mmol) in THF (2.00 mL) at -78°C under N_2 , and the mixture was stirred at 0°C for 10 min. A solution of (**Z**)-**8c** (127 mg, 0.300 mmol) in THF (1.00 mL) was added dropwise to the solution of the organocopper reagent at -78°C . After stirring at 0°C for 2.0 h, the reaction was quenched by addition of NH_4Cl -28% NH_3 aqueous solution with additional stirring at room temperature for 30 min. The mixture was extracted with Et_2O , and the extract was washed with brine and dried over MgSO_4 . Concentration under reduced pressure followed by flash column chromatography over silica gel with *n*-hexane-EtOAc (10:1) gave the title compound **10c** as a light yellow oil (116 mg, 96%, >20:1 dr): $[\alpha]_{\text{D}} = -9.16$ (c 1.00, CHCl_3); IR (ATR) ν 1733 (CO), 1366 (NHSO_2), 1128 (NHSO_2) cm^{-1} ; ^1H NMR (500 MHz, CDCl_3) δ 1.20 (t, $J = 7.10$ Hz, 3H), 1.21 (d, $J = 7.15$ Hz, 3H), 1.28 (s, 9H), 3.00 (dd, $J = 13.6$, 7.40 Hz, 1H), 3.08 (dd, $J = 13.6$, 6.50 Hz, 1H), 3.52 (dq, $J = 9.10$, 7.15 Hz, 1H), 4.06 (q, $J = 7.10$ Hz, 2H), 4.15 (d, $J = 9.4$ Hz, 1H), 4.33 (ddd, $J = 9.40$, 7.40, 6.50 Hz, 1H), 5.64 (d, $J = 9.10$ Hz, 1H), 7.17–7.30 (m, 5H); ^{13}C NMR (125 MHz, CDCl_3) δ 14.3, 17.0, 24.1 (3C), 39.7, 41.2, 60.1, 61.0, 61.7, 127.0, 128.0, 129.0 (2C), 130.0 (2C), 135.0, 136.0, 173.0; HRMS (ESI), m/z calcd for $\text{C}_{19}\text{H}_{29}\text{ClNO}_4\text{S}$ $[\text{M}+\text{H}]^+$ 402.1500, found 402.1501.

4.3.10. Ethyl (2*R*,5*S*,*Z*)-2-benzyl-5-((tert-butoxycarbonyl)amino)-4-chloro-6-methylhept-3-enoate (**11**)

Anisole (326 μL , 3.00 mmol) and AlCl_3 (333 mg, 2.50 mmol) were added to the solution of ester **7c** (430 mg, 1.00 mmol) in CH_2Cl_2 (10.0 mL) under N_2 . After stirring for 2.0 h at room temperature, the reaction was quenched with NH_3 aqueous solution and saturated aqueous Rochelle salt solution then extracted with CH_2Cl_2 . The extract was washed with brine and dried over MgSO_4 . The reaction mixture was concentrated under reduced pressure to give the deprotected amine, which was used immediately in the

next step without purification. Boc₂O (262 mg, 1.20 mmol) and DIPEA (261 μ L, 1.50 mmol) were added to the solution of the amine in CH₂Cl₂ (10.0 mL). After stirring for 2.0 h at room temperature, the reaction was quenched with saturated aqueous NH₄Cl solution and extracted with CH₂Cl₂. The extract was washed with brine and dried over MgSO₄. Concentration under reduced pressure followed by flash chromatography over silica gel with *n*-hexane-EtOAc (12:1) gave the title compound **11** (400 mg, 98%) as a white solid: $[\alpha]_D = -61.1$ (c 1.01, CHCl₃); IR (ATR) ν 3250 (NH), 1729 (CO) cm⁻¹; ¹H NMR (500 MHz, CDCl₃) δ 0.852 (d, *J* = 6.6 Hz, 3H), 0.919 (d, *J* = 6.6 Hz, 3H), 1.15 (t, *J* = 7.1 Hz, 3H), 1.45 (s, 9H), 1.82 (dq, *J* = 9.0, 6.6 Hz, 1H), 2.89 (dd, *J* = 13.6, 7.2 Hz, 1H), 3.07 (dd, *J* = 13.6, 7.5 Hz, 1H), 3.79 (ddd, *J* = 9.1, 7.5, 7.2 Hz, 1H), 3.94 (t, *J* = 9.0 Hz, 1H), 4.07 (q, *J* = 7.1 Hz, 2H), 4.71 (d, *J* = 9.0 Hz, 1H), 5.83 (d, *J* = 9.1 Hz, 1H), 7.17–7.25 (m, 5H); ¹³C NMR (125 MHz, CDCl₃) δ 14.2, 18.7, 19.7, 28.5 (2C), 30.6, 38.3, 41.2, 61.0, 62.9, 79.7, 125.3, 126.6, 128.4 (2C), 129.3 (2C), 136.8, 138.2, 155.2, 172.63; HRMS (ESI), *m/z* calcd for C₂₂H₃₃ClNO₄ [M+H]⁺ 410.2093, found 410.2096.

4.3.11. (2*R*,5*S*,*Z*)-2-Benzyl-5-((*tert*-butoxycarbonyl)amino)-4-chloro-6-methylhept-3-enoic acid (**12**)

A solution of DIBAL-H in toluene (1.00 M, 604 μ L, 604 μ mol) was added dropwise to a solution of **11** (225 mg, 549 μ mol) in CH₂Cl₂ (10.0 mL) at -78 °C under N₂, and the mixture was stirred for 1.0 h at -78 °C. The reaction was quenched with saturated aqueous Rochelle salt solution and extracted with Et₂O. The extract was washed with brine and dried over MgSO₄. The reaction mixture was concentration under reduced pressure to give an aldehyde, which was used immediately in the next step without purification. NaH₂PO₄ (329 mg, 2.74 mmol) in H₂O (5.49 mL), 2-methyl-2-butene (583 μ L, 5.49 mmol) and NaClO₂ (93.1 mg, 823 μ mol, >80.0% purity) were added to a solution of the aldehyde in *t*-BuOH (5.49 mL). After stirring for 2.0 h at room temperature, the reaction mixture was extracted with EtOAc, and the extract was washed with brine and dried over MgSO₄. Concentration under reduced pressure followed by flash chromatography over silica gel with *n*-hexane-EtOAc (2:1) gave the title compound **12** as a white solid (195 mg, 93%); $[\alpha]_D = -41.6$ (c 1.06, CHCl₃); IR (ATR) ν 3300 (NH), 1709 (CO) cm⁻¹; ¹H NMR (500 MHz, CDCl₃) δ 0.854 (d, *J* = 6.6 Hz, 3H), 0.910 (d, *J* = 6.6 Hz, 3H), 1.46 (s, 9H), 1.82 (dq, *J* = 9.0, 6.6 Hz, 1H), 2.91 (dd, *J* = 13.6, 7.4 Hz, 1H), 3.13 (dd, *J* = 13.6, 7.1 Hz, 1H), 3.84 (ddd, *J* = 9.0, 7.4 Hz, 1H), 3.94 (t, *J* = 9.0 Hz, 1H), 4.71 (d, *J* = 9.0 Hz, 1H), 5.83 (d, *J* = 9.0 Hz, 1H), 7.18–7.27 (m, 5H); ¹³C NMR (125 MHz, CDCl₃) δ 18.6, 19.7 (2C), 28.5 (3C), 30.5, 38.1, 46.7, 62.9, 79.9, 124.6, 126.8, 128.5, 128.6, 129.33 (2C), 137.5, 137.9, 155.2; HRMS (ESI), *m/z* calcd for C₂₀H₂₉ClNO₄ [M+H]⁺ 382.1780, found 382.1781.

4.3.12. Ethyl (2*R*,5*S*,*Z*)-5-(((9*H*-fluoren-9-yl)methoxy)carbonyl)amino)-2-benzyl-4-chloro-6-methylhept-3-enoate (**13**)

Anisole (326 μ L, 3.00 mmol) and AlCl₃ (333 mg, 2.50 mmol) were added to the solution of the ester **7c** (430 mg, 1.00 mmol) in CH₂Cl₂ (10.0 mL) under N₂. After stirring for 2.0 h at room temperature, the reaction was quenched with NH₃ aqueous, saturated aqueous Rochelle salt solution and extracted with CH₂Cl₂. The extract was washed with brine and dried over MgSO₄. The reaction mixture was concentrated under reduced pressure to give the deprotected amine, which was used immediately in the next step without purification. FmocCl (285 mg, 1.10 mmol) and DIPEA (200 μ L, 1.15 mmol) were added to a solution of the amine in CH₂Cl₂ (10.0 mL). After stirring for 2.0 h at room temperature, the reaction was quenched with saturated aqueous NH₄Cl solution and extracted with CH₂Cl₂. The extract was washed with brine and dried over MgSO₄. Concentration under reduced pressure followed by flash chromatography over silica gel with *n*-hexane-EtOAc (10:1) gave

the title compound **13** (516 mg, 97%) as a white solid: $[\alpha]_D = -36.5$ (c 1.00, CHCl₃); IR (ATR) ν 3338 (NH), 1726 (CO) cm⁻¹; ¹H NMR (500 MHz, CDCl₃) δ 0.8625 (d, *J* = 6.6 Hz, 3H), 0.937 (d, *J* = 6.6 Hz, 3H), 1.15 (t, *J* = 7.2 Hz, 3H), 1.84 (dq, *J* = 9.6, 6.6 Hz, 1H), 2.87 (dd, *J* = 13.5, 7.5 Hz, 1H), 3.07 (dd, *J* = 13.5, 7.3 Hz, 1H), 3.78 (ddd, *J* = 9.2, 7.5, 7.3 Hz, 1H), 3.98 (dd, *J* = 9.6, 9.5 Hz, 1H), 4.08 (q, *J* = 7.2 Hz, 2H), 4.23 (t, *J* = 7.0 Hz, 1H), 4.37 (dd, *J* = 10.7, 7.0 Hz, 1H), 4.48 (dd, *J* = 10.7, 7.0 Hz, 1H), 4.93 (d, *J* = 9.5 Hz, 1H), 5.85 (d, *J* = 9.2 Hz, 1H), 7.00–7.20 (m, 5H), 7.31–7.34 (m, 2H), 7.40–7.43 (m, 2H), 7.59–7.61 (m, 2H), 7.77–7.79 (m, 2H); ¹³C NMR (125 MHz, CDCl₃) δ 14.2, 18.9, 19.7, 30.5, 38.2, 47.2, 47.4, 61.0, 63.6, 66.9, 120.2, 125.1, 125.2, 125.9 (2C), 126.6 (2C), 127.2 (2C), 127.9 (2C), 128.4 (2C), 129.3 (2C), 136.3, 138.0, 141.5, 144.0, 144.1, 155.7, 172.5; HRMS (ESI), *m/z* calcd for C₃₂H₃₅ClNO₄ [M+H]⁺ 532.2249, found 532.2252.

4.3.13. (2*R*,5*S*,*Z*)-5-(((9*H*-fluoren-9-yl)methoxy)carbonyl)amino)-2-benzyl-4-chloro-6-methylhept-3-enoic acid (**14**)

A solution of DIBAL-H in toluene (1.00 M, 220 μ L, 220 μ mol) was added dropwise to a solution of **13** (106 mg, 200 μ mol) in CH₂Cl₂ (2.00 mL) at -78 °C under nitrogen, and the mixture was stirred for 1.0 h at -78 °C. The reaction was quenched with saturated aqueous Rochelle salt solution and extracted with Et₂O. The extract was washed with brine and dried over MgSO₄. The reaction mixture was concentrated under reduced pressure to give the aldehyde, which was used immediately in next step without purification. NaH₂PO₄ (120 mg, 1.00 mmol) in H₂O (2.00 mL), 2-methyl-2-butene (213 μ L, 2.00 mmol) and NaClO₂ (33.9 mg, 300 μ mol, >80.0% purity) were added to the solution of the aldehyde in *t*-BuOH (2.00 mL). After stirring for 2.0 h at room temperature, the reaction mixture was extracted with EtOAc, and the extract was washed with brine and dried over MgSO₄. Concentration under reduced pressure followed by flash chromatography over silica gel with *n*-hexane-EtOAc (2:1) gave the title compound **14** as a white solid (96.8 mg, 96%); $[\alpha]_D = -45.6$ (c 1.02, CHCl₃); IR (ATR) ν 3376 (NH), 1714 (CO) cm⁻¹; ¹H NMR (500 MHz, CDCl₃) δ 0.856 (d, *J* = 6.6 Hz, 3H), 0.918 (d, *J* = 6.6 Hz, 3H), 1.84 (dq, *J* = 9.3, 6.6 Hz, 1H), 2.88 (dd, *J* = 13.6, 7.0 Hz, 1H), 3.12 (dd, *J* = 13.6, 6.9 Hz, 1H), 3.83 (ddd, *J* = 9.0, 7.0, 6.9 Hz, 1H), 3.98 (dd, *J* = 9.4, 9.3 Hz, 1H), 4.23 (t, *J* = 7.0 Hz, 1H), 4.38 (dd, *J* = 10.6, 7.0 Hz, 1H), 4.49 (dd, *J* = 10.6, 7.0 Hz, 1H), 4.94 (d, *J* = 9.4 Hz, 1H), 5.84 (d, *J* = 9.0 Hz, 1H), 7.16–7.18 (m, 5H), 7.31–7.34 (m, 2H), 7.39–7.42 (m, 2H), 7.58–7.61 (m, 2H), 7.77–7.79 (m, 2H); ¹³C NMR (125 MHz, CDCl₃) δ 18.9, 19.7, 30.5, 38.0, 46.7, 47.4, 63.6, 66.9, 120.2 (2C), 125.1, 125.2, 125.3, 126.7 (2C), 127.2 (2C), 127.9 (2C), 128.4 (2C), 129.3 (2C), 137.0, 137.6, 137.7, 141.5, 143.9, 144.0, 155.7; HRMS (ESI), *m/z* calcd for C₃₀H₃₁ClNO₄ [M+H]⁺ 504.1936, found 504.1937.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.tet.2017.06.003>.

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